

Enterprise Server Deployment and Operating System Installation

Installation Guide

Week 3 Portfolio Project

Course: ITEP 414 – System Administration and Maintenance

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Project Overview

This guide documents the deployment of an Ubuntu Server virtual machine for a startup company. It covers installation, initial configuration, verification, and documentation so another system administrator can reproduce the setup.

Objectives

- Install Ubuntu Server in a virtual machine.
- Configure hostname, networking, storage, and SSH.
- Verify that the server is functioning correctly.
- Document the deployment process professionally.

Software Requirements

Software	Purpose
Oracle VirtualBox or VMware	Virtualization
Ubuntu Server LTS ISO	Operating System
Windows Server Evaluation ISO	Bring-home activity

Virtual Machine Specifications

Component	Specification
Name	Ubuntu-Server-Week03
RAM	4 GB
CPU	2 Virtual Processors
Storage	40 GB
Network	NAT
Hostname	server01

Ubuntu Server Installation Procedure

1. Create a new virtual machine named Ubuntu-Server-Week03.

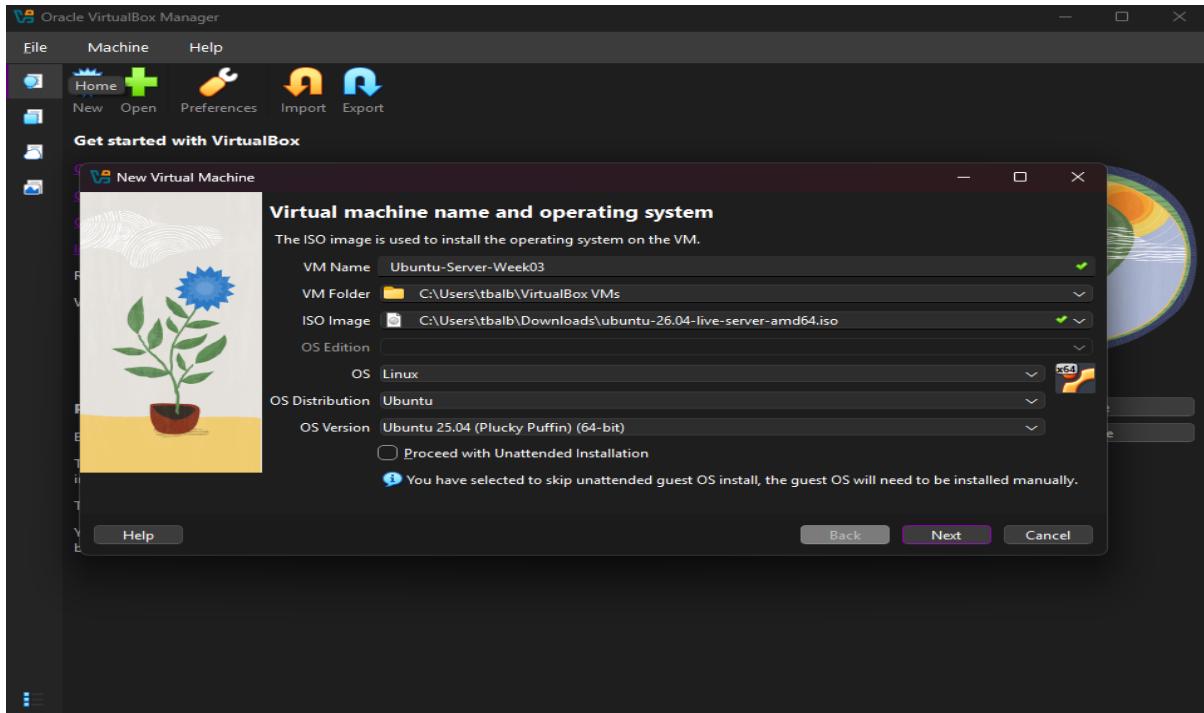


Figure 1. Ubuntu Server Creation

2. Allocate 4 GB RAM and 2 virtual processors.

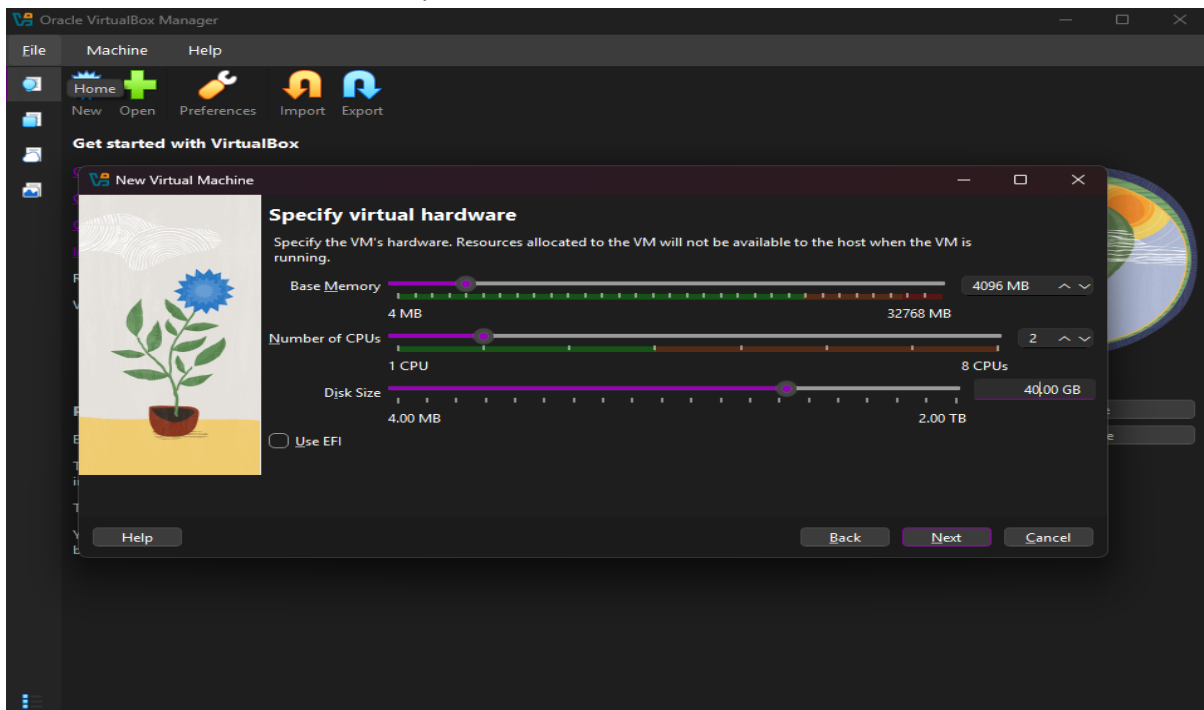


Figure 2. Ubuntu Server Hardware Configuration

3. Attach the Ubuntu Server LTS ISO and boot the VM.

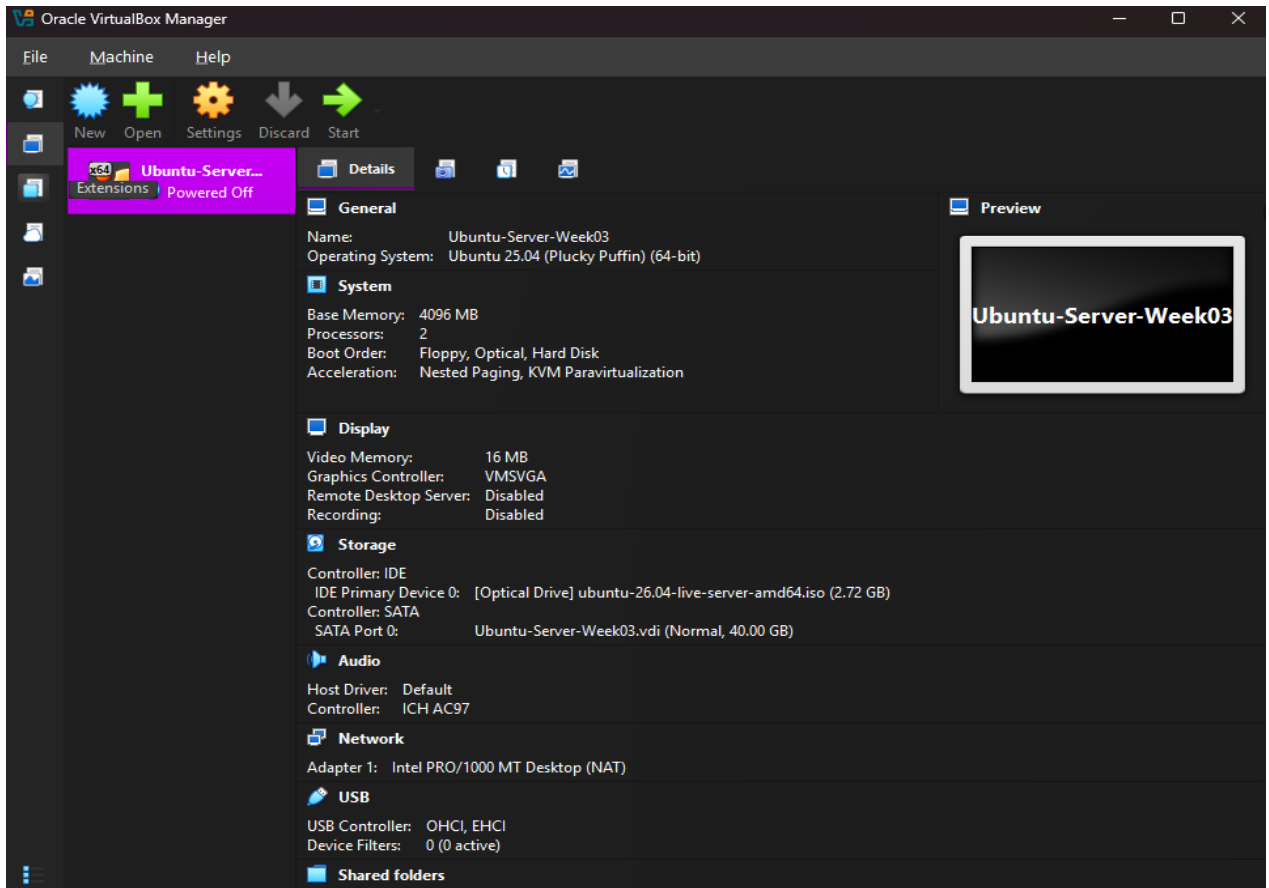


Figure 3. Ubuntu Server Summary

4. Select English language and English (US) keyboard layout.

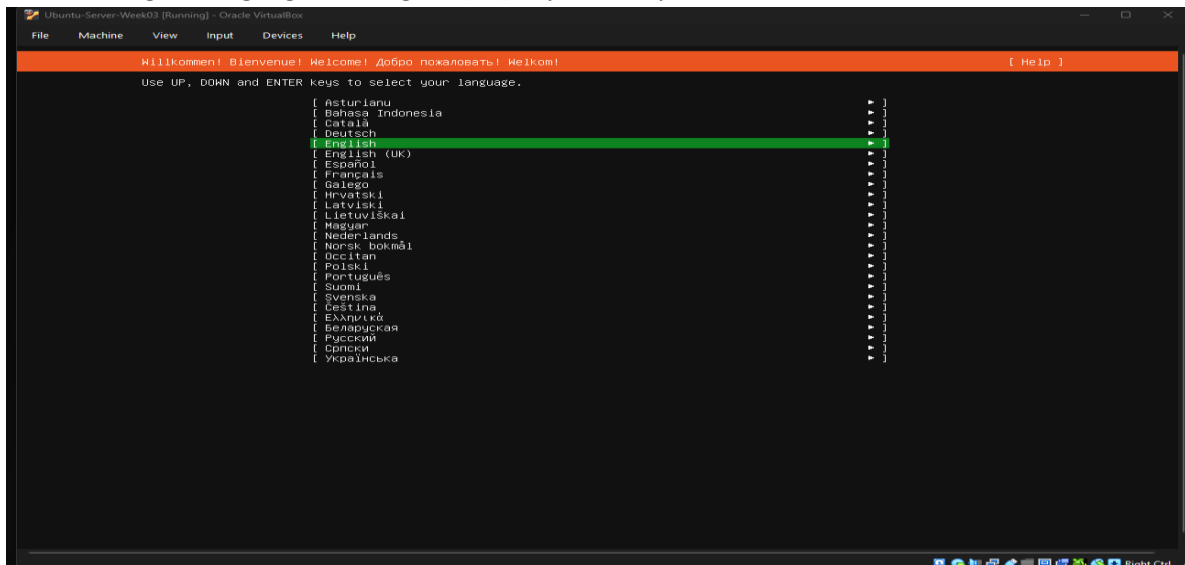


Figure 4.1. Ubuntu Server Language Selection

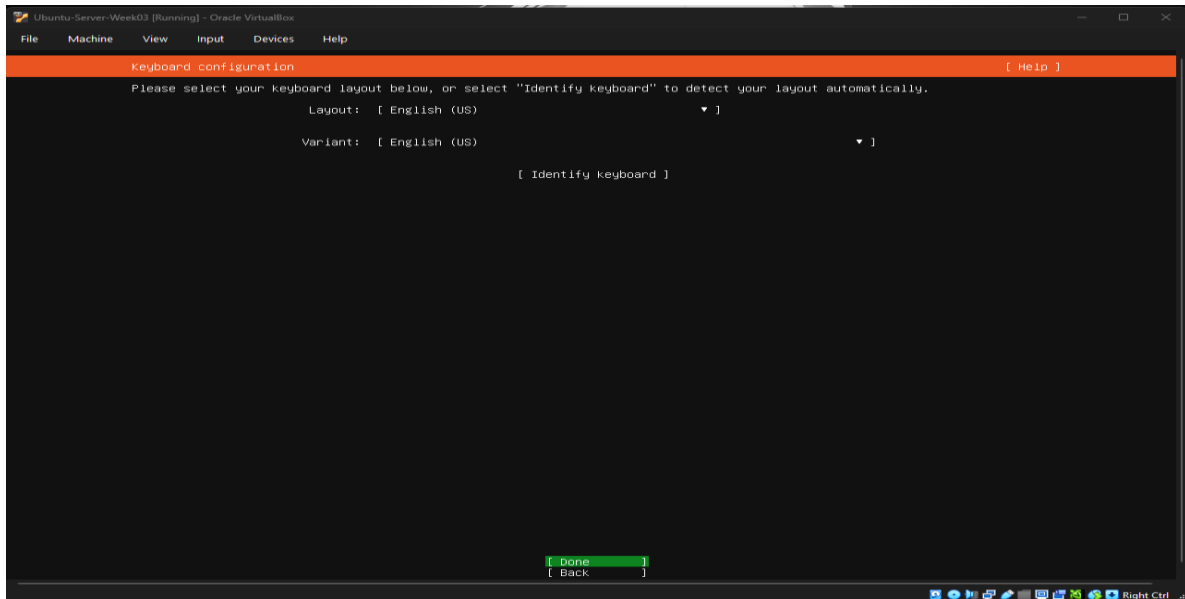


Figure 4.2. Ubuntu Server Keyboard Layout Selection

5. Accept DHCP for network configuration.

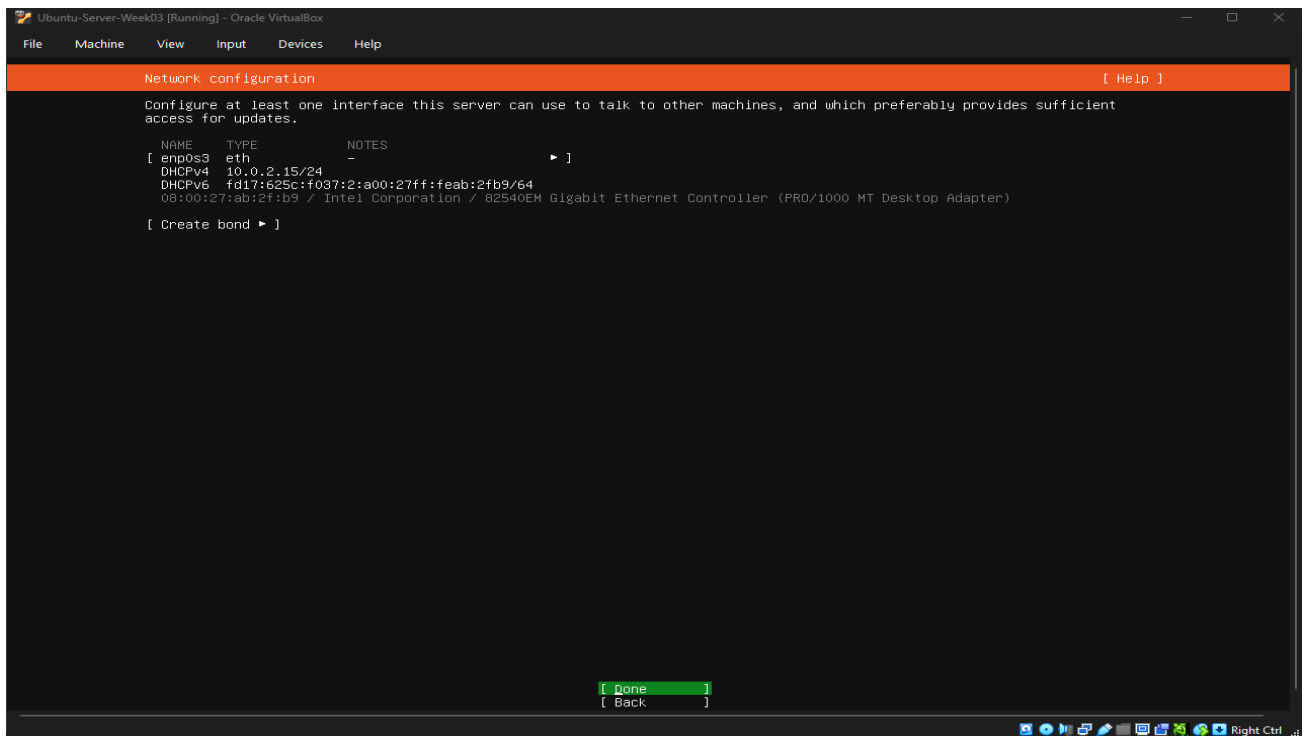


Figure 5. Ubuntu Server Network Configuration

6. Set the hostname to server01.

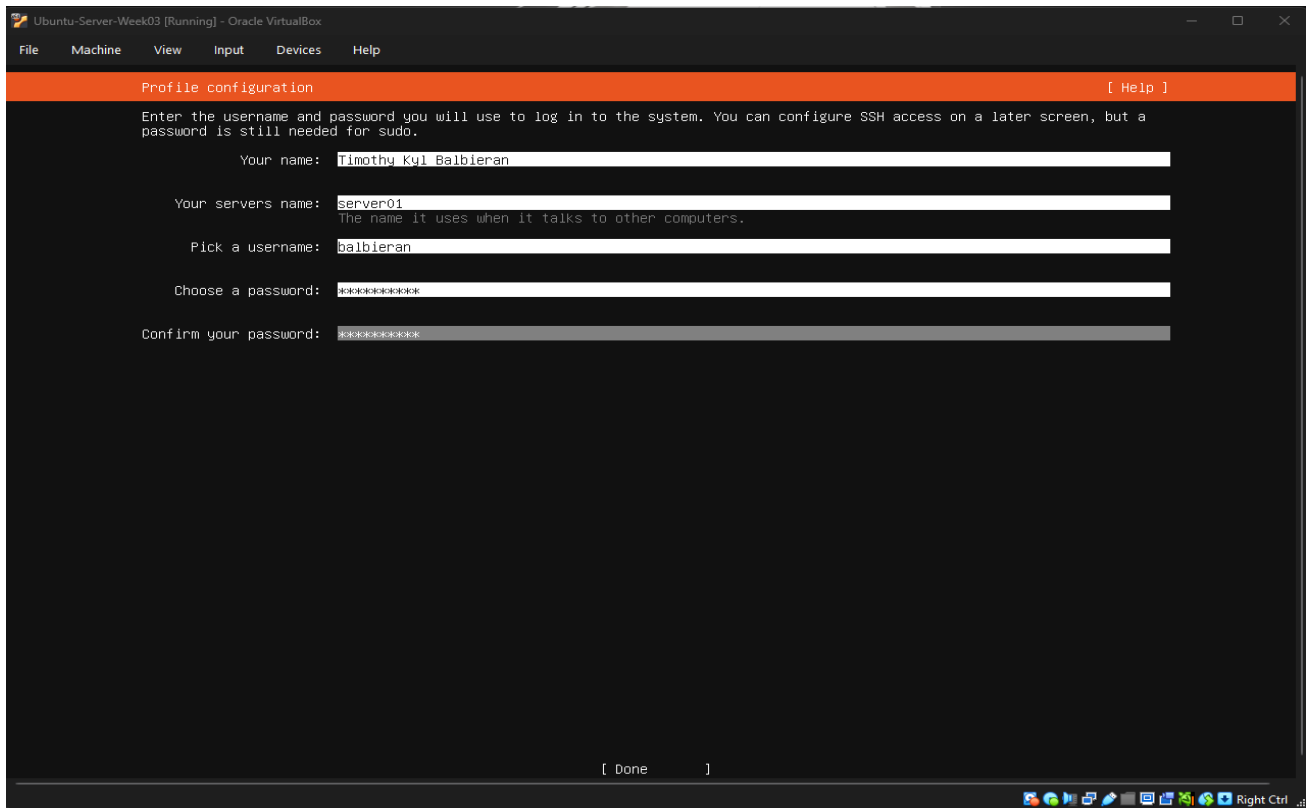


Figure 6. Ubuntu Server Profile Configuration

7. Create a non-root administrator account.

8. Use Guided – Use Entire Disk for storage.

9. Enable Install OpenSSH Server

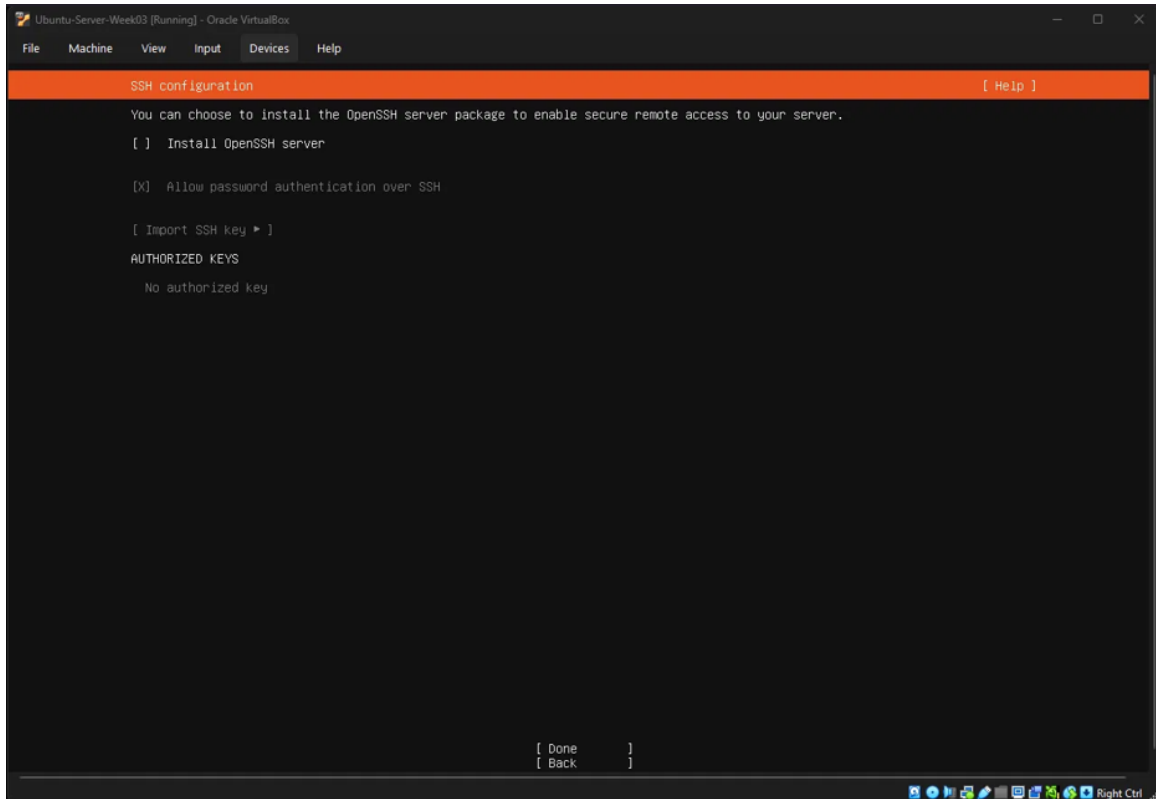


Figure 7. Ubuntu Server SSH Configuration

10. Complete the installation and restart the virtual machine.

Initial Configuration

Username: balbieran

Network Interface: enp0s3

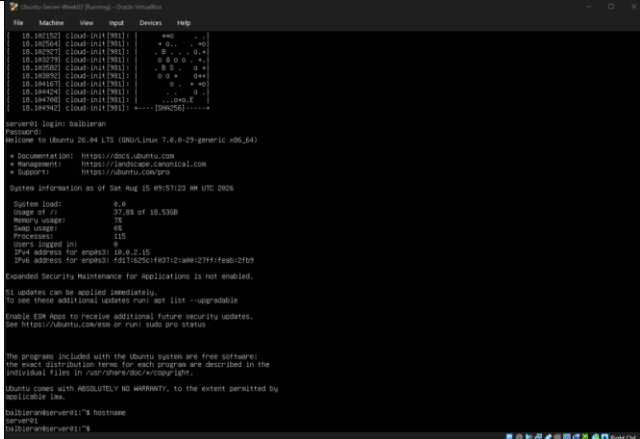
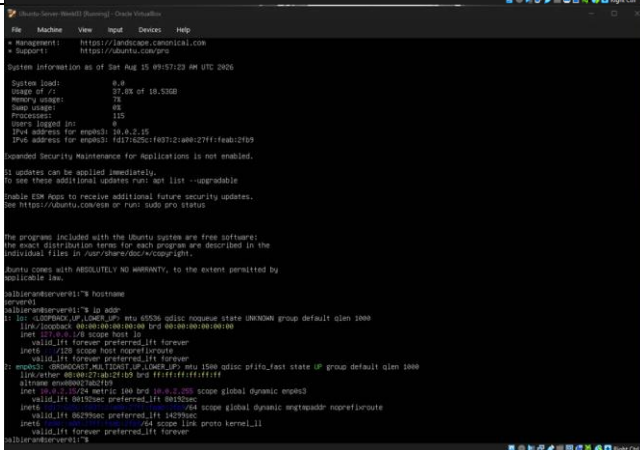
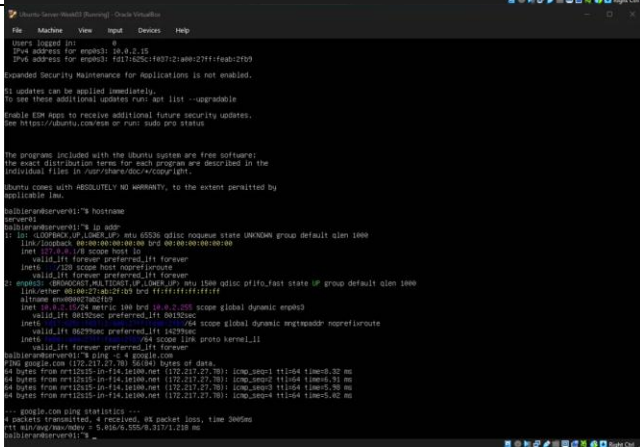
Assigned IPv4 Address: 10.0.2.15/24

Network Configuration: DHCP

Network Mode: NATs

File System: ext4

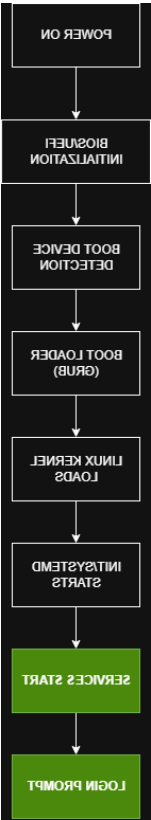
Server Verification

Command	Purpose	Screenshot
hostname	Verify hostname	 A terminal window titled 'balbierzserver01' showing system information for Ubuntu 20.04 LTS. The output of the 'hostname' command is 'balbierzserver01'.
ip addr	View IP address	 A terminal window titled 'balbierzserver01' showing the output of the 'ip addr' command. It displays details for the 'eth0' interface, including the IP address '10.0.0.1'.
ping -c 4 google.com	Test internet connectivity	 A terminal window titled 'balbierzserver01' showing the output of the 'ping -c 4 google.com' command. It displays the IP address of google.com (172.217.27.70) and the results of four successful ping attempts.

	attack the boot process.	and OS haven't been tampered with before letting the system boot.
Speed	Boots slower since it loads hardware one piece at a time in an old 16-bit mode.	Boots faster because it can initialize hardware at the same time and runs in 32-bit or 64-bit mode.
Advantages	Simple and works on old hardware.	Faster boots, supports big drives, more secure, and has a nicer setup menu you can navigate with a mouse.
Disadvantages	Slow, can't handle large drives, no security features, and the interface is just plain text.	A bit more complex, and old operating systems might not even support it.
Modern Usage	Barely used anymore, mostly just on old machines or legacy systems.	Basically, the standard now on all new computers and servers, including the VM I set up for this project.

Ubuntu Boot Process Flowchart

Power On → BIOS/UEFI → Boot Device Detection → GRUB Boot Loader → Linux Kernel →
 init/systemd → Services Start → Login Prompt



Windows Server Installation Summary

Summarize the Windows Server installation after completing the bring-home activity.

Operating System Comparison

Category	Windows	Ubuntu	Rocky
Licensing	Paid, requires a product license (or evaluation copy for testing). Licensing is per-core or per-CAL	Free and open source. No licensing cost, though paid support (Ubuntu Pro) is optional.	Free and open source, built as a downstream rebuild of Red Hat Enterprise Linux (RHEL) with no

	depending on edition.		license fee.
Interface	Has a full graphical desktop by default (though a Server Core, command-line-only option exists too). Very approachable for people coming from regular Windows.	Command-line only by default (no GUI on Server edition). Managed through the terminal or remotely through SSH.	Same as Ubuntu, command-line focused, though a GUI can be installed if needed.
Package Management	Uses PowerShell and the Windows Installer/MSI system, plus Windows Update for patches.	Uses apt for installing and updating software, which is simple and has a huge repository.	Uses dnf (or yum), the same package manager RHEL uses, which is known for being stable and enterprise focused.
Security	Built-in Windows Defender, Windows Firewall, and Active Directory integration for enterprise security policies.	Security updates through apt, AppArmor for access control, and a strong reputation for quick patching.	Security-Enhanced Linux (SELinux) by default, plus frequent updates since it tracks RHEL closely.
Performance	Heavier on system resources because of the GUI and background services, even in Server Core mode.	Lightweight runs well on minimal hardware since there's no GUI overhead.	Similarly lightweight, performance is close to Ubuntu since both are Linux based.
Typical Enterprise Use Cases	Active Directory domains, .NET applications, Microsoft Exchange, SQL Server, environments already built around Microsoft tools.	Web hosting, cloud infrastructure, Docker/Kubernetes, DevOps pipelines, startups and cloud-native companies.	Enterprise environments that need RHEL-like stability without paying for RHEL, common in banking, government, and long-term production systems.
Advantages	Easy to use for Windows-familiar admins, strong enterprise support, tight integration with Microsoft products.	Free, huge community support, frequent updates, works great for cloud and container workloads.	Very stable, long support lifecycle, good for organizations that need predictability over the newest features.
Disadvantages	Expensive licensing, heavier resource	Not ideal for organizations already	Smaller community than Ubuntu, slower

	usage, more attack surface due to the GUI.	built around Microsoft software.	to get bleeding-edge software versions.
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Problems Encountered

- During `sudo apt upgrade`, the package `linux-firmware-amd-misc` repeatedly failed to download from the Philippines Ubuntu mirror (403 Forbidden). This did not affect server functionality, since the package is optional firmware not required for the VM.
- The "Install OpenSSH server" checkbox during installation did not apply correctly. Running `systemctl status ssh` after logging in returned "Unit `ssh.service` could not be found."

Solutions Implemented

- Re-ran `sudo apt update` and `sudo apt full-upgrade -y --fix-missing` to complete the update while skipping the unavailable firmware package.
- Manually installed and enabled SSH post-installation with `sudo apt install openssh-server -y`, followed by `sudo systemctl enable ssh` and `sudo systemctl start ssh`. Verified with `systemctl status ssh`, which then showed active (running).

Lessons Learned

Installing Ubuntu Server demonstrated the importance of careful system configuration, documentation, and verification before deploying an enterprise server.

References

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<https://documentation.ubuntu.com/server/>

Microsoft. (2024). *Windows Server evaluations*. <https://www.microsoft.com/en-us/evalcenter/download-windows-server-2025>

Red Hat, Inc. (2024). *What is Rocky Linux?* <https://www.redhat.com/en/topics/linux/what-is-rocky-linux>

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